DSP · L05

DTFT and its properties

Cheat sheet · Day 06 · Thu 29 Oct · pack D3

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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DTFT (L05)  X(w) = sum x(n) e^{-jwn};  x(n) = (1/2pi) ∫_{2pi} X(w) e^{jwn} dw
 converges if sum|x|<inf or sum|x|^2<inf;  X(w+2pi k)=X(w)  -> reduce w to (-pi,pi]
DTFS: x(n)=sum_<N> c_k e^{jk w0 n}, c_k=(1/N) sum_<N> x(n)e^{-jk w0 n}, w0=2pi/N, c_
      (1/N) sum|x|^2 = sum|c_k|^2
SHORTCUTS (mark n=0 at the arrow first!)
 X(0)=sum x    X(pi)=sum (-1)^n x    ∫X dw = 2pi x(0)    ∫X e^{jwn0} dw = 2pi x(n0)
 ∫|X|^2 dw = 2pi sum|x|^2    X'(0) = -j sum n x(n)    angle of negative real = ±pi
PAIRS: delta(n-n0) -> e^{-jwn0};  a^n u(n) -> 1/(1-a e^{-jw}) |a|<1
 u(n)-u(n-M) -> (1-e^{-jwM})/(1-e^{-jw}) = e^{-jw(M-1)/2} sin(Mw/2)/sin(w/2)
 b^n u(-n) -> 1/(1-b^{-1}e^{jw}) |b|>1;   a^n u(n+n0) -> a^{-n0} e^{jwn0}/(1-a e^{-jv
 n a^n u(n) -> a e^{-jw}/(1-a e^{-jw})^2;  (n+1)a^n u(n) -> 1/(1-a e^{-jw})^2
 a^|n| -> (1-a^2)/(1-2a cos w + a^2);  cos(w0 n) -> pi[δ(w-w0)+δ(w+w0)] (period 2pi)
PROPERTIES: x(n-n0)->e^{-jwn0}X ; e^{jw0 n}x->X(w-w0) ; (-1)^n x -> X(w-pi)
 x*(n)->X*(-w); real x: |X| even, phase odd ; x(-n)->X(-w)
 x(n)-x(n-1)->(1-e^{-jw})X ; sum_{-inf}^n x -> X/(1-e^{-jw}) + pi X(0) sum δ(w-2pi k)
 x_(k)(n) -> X(kw) (compress, images) ; n x(n) -> j dX/dw
 x*h -> XH ;  x1 x2 -> (1/2pi) ∫_{2pi} X1(λ)X2(w-λ)dλ (periodic conv.)
SAMPLING: X(w)=(1/T) sum_k Xa((w-2pi k)Fs), w = 2pi F/Fs ; no alias iff Fs > 2 F_M
 recover: ideal LPF gain T, cutoff between W_M and Ws - W_M
METHOD: finite -> term by term; right-sided -> shift index to start at 0, geometric
 series; two-sided -> split n>=0 and n<0; ALWAYS check X(0)=sum x.
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